

12. Battery Backup Function

12.1 Overview

When the voltage at the VCC pin is dropped, power can be supplied to the realtime clock (RTC) and sub-clock oscillator from the dedicated battery backup power pin (VBATT pin). When the voltage drop at the VCC pin is detected, connection to power is switched to the VBATT pin.

Figure 12.1 shows the configuration of the battery backup function.

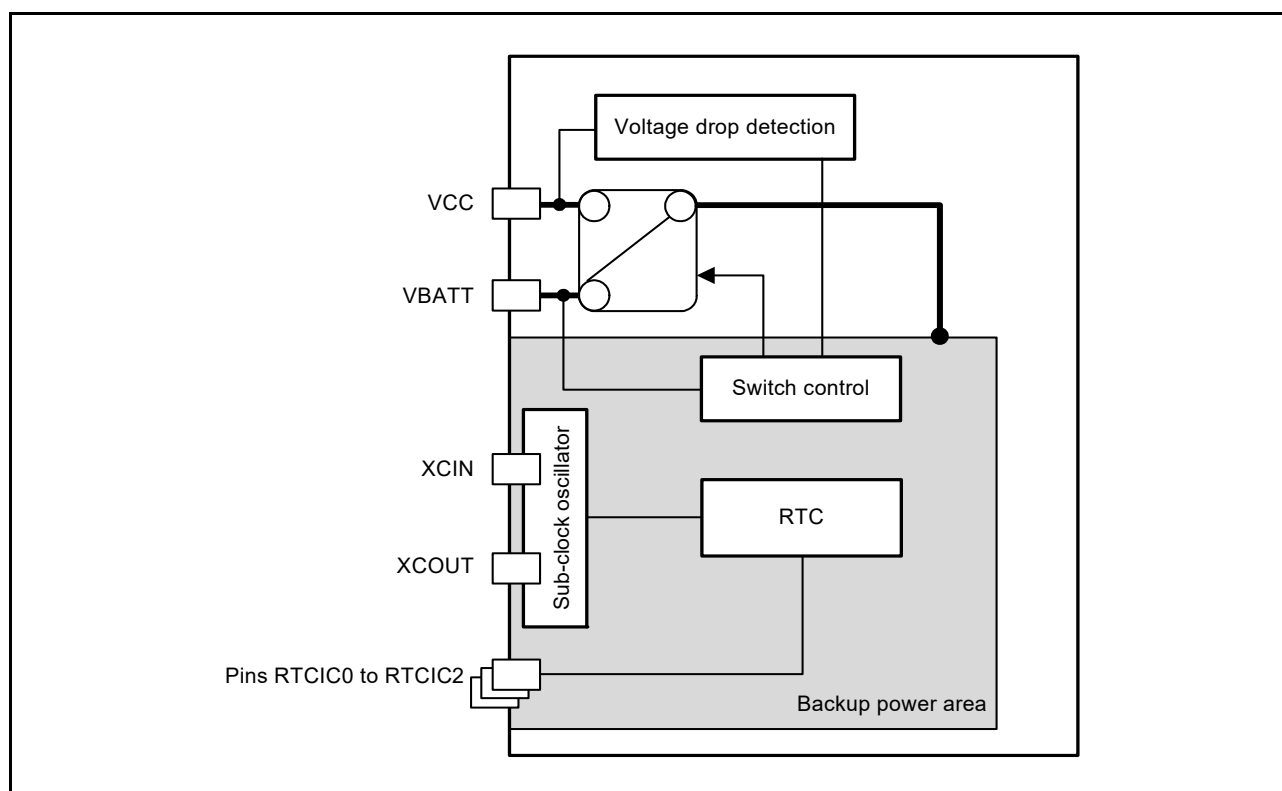


Figure 12.1 Configuration of Battery Backup Function

12.2 Operation

12.2.1 Battery Backup Function

When the voltage at the VCC pin is dropped, power can be supplied to the RTC and sub-clock oscillator from the VBATT pin. When the power supply reduction from the VCC pin is detected, connection to power is switched to the power supply from the VBATT pin. The power supply from the VCC pin is resumed when the voltage at the VCC pin exceeds VDET BATT while the RTC is operating on the power supply from the VBATT pin. This power supply change does not affect the RTC operation. When the voltage level at the VBATT pin voltage falls below the operation guaranteed voltage, operation of the RTC cannot be guaranteed.

In addition, the battery backup function should be used after the voltage monitoring 0 reset is enabled.

The power is supplied to the following modules from the VBATT pin.

- RTC
- Sub-clock oscillator (including XCIN and XCOU T pins)
- P30, P31, and P32 input pins

Figure 12.2 shows the operation for switching to the battery backup function.

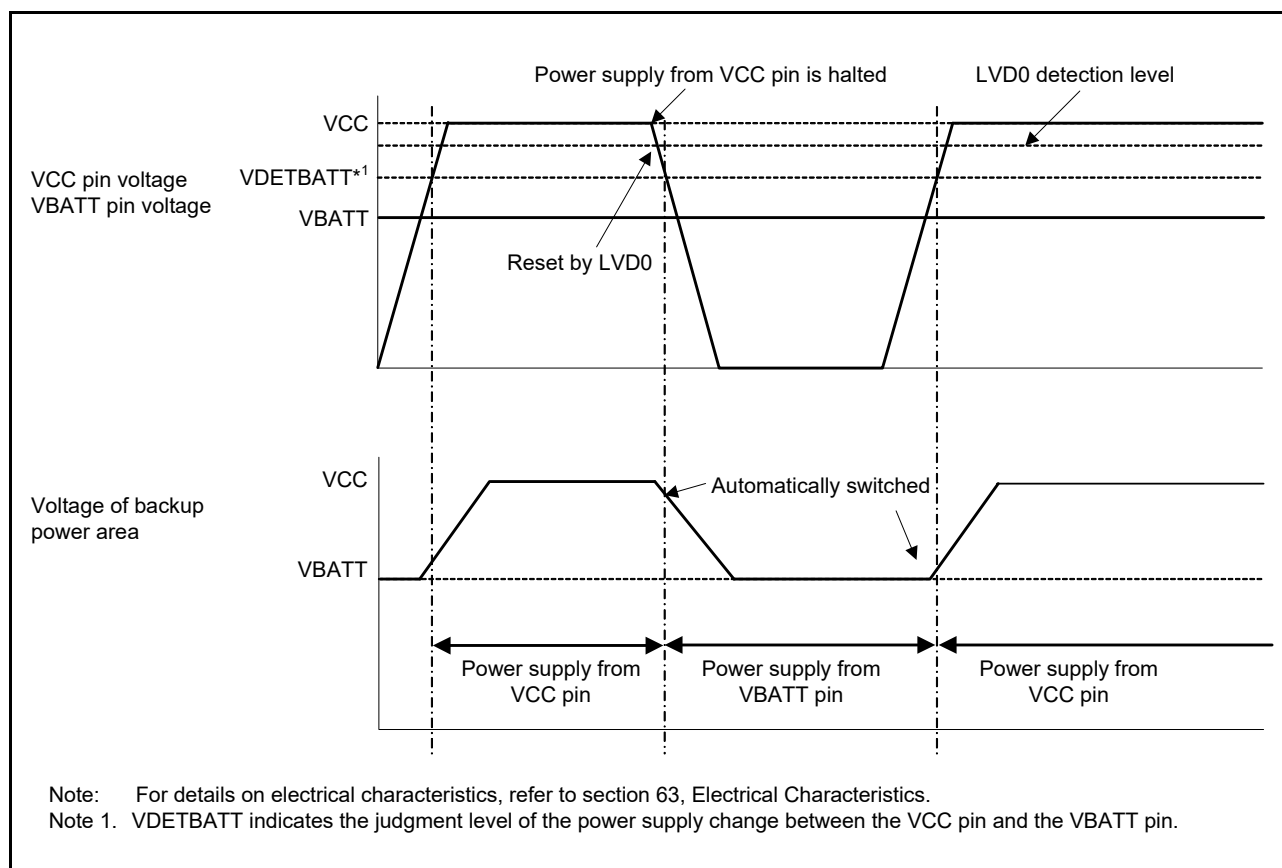


Figure 12.2 Operation for Switching to Battery Backup Function

12.3 Usage Notes

1. When the VBATT pin is not in use, connect the VBATT pin to the VCC pin.
2. When the voltage level at the VBATT is lower than the guaranteed operation range, operation of the sub-clock and RTC cannot be guaranteed. The RTC must be initialized to restart power supply after the VBATT pin falls below the operation guaranteed voltage.
3. Writing to the RTC registers should be performed while power is being supplied from the VCC pin.
4. When VCC is higher than VDET_{BATT}, the VCC pin and VBATT pin are not connected by means of circuitry. When VCC is lower than VDET_{BATT} and the switch is connected to the VBATT pin, if the voltage at the VBATT pin becomes lower than (the range from VCC to 0.6 V), current may flow into the VBATT pin through the parasitic diode between the VCC and VBATT pins.
5. During RTC operation using the voltage from the VBATT pin, I/O ports (P30, P31, and P32) within the backup power supply area can only be used as time capture event input pins for the RTC.